

Structural Information as the Missing Component in the Radio–Star-Formation-Rate Relation

JASON S. MAKECHEMU,^{1,2} JAMES O. CHIBUEZE,^{2,3} AND BROOKE D. SIMMONS¹

¹*Physics Department, Lancaster University, Lancaster LA1 4YB, UK*

²*Department of Mathematical Sciences, University of South Africa, Pretoria 0003, South Africa*

³*Department of Physics and Astronomy, University of Nigeria, Nsukka 410001, Nigeria*

ABSTRACT

The infrared-radio correlation carries $\sigma \sim 0.2$ – 0.3 dex of scatter whose physical origin has remained unidentified. We quantify galaxy structure as the dominant hidden variable, confirmed independently by two wholly distinct morphological measurement systems applied to $N = 4843$ VLA-COSMOS 3 GHz sources (median $z = 0.91$). An eight-rung information ladder using a multilayer perceptron (MLP) shows that non-parametric morphology (Gini, concentration, asymmetry) from HST/ACS delivers $\Delta R^2 = +0.127$, a gain $4.3\times$ larger than all other variables combined, raising cumulative R^2 from 0.50 to 0.62. A redshift-sensitivity check confirms that $\Delta R^2 = 0.038$ survives redshift control, remaining the single largest per-variable improvement available. The Galaxy Zoo Hubble citizen-science catalogue independently replicates the signal. The de-biased featured/disk vote fraction alone delivers $\Delta R^2 = +0.156$, and the combined AUC from all Galaxy Zoo Hubble features, 0.697 ± 0.009 , is indistinguishable from the HST non-parametric AUC, 0.693 ± 0.009 . A redshift-sensitivity check on the GZ Hubble ladder yields a retention fraction of 0.50 — higher than the Zurich value of 0.30 because volunteer debiasing has already removed the dominant redshift trend from the vote fractions. The redshift-controlled GZ Hubble residual ($\Delta R^2 = 0.078$) exceeds the Zurich residual ($\Delta R^2 = 0.038$), making it the largest per-variable structural gain after redshift control.

Matched-sample tests at fixed $(\log(M_*/M_\odot), z, \log L_{\text{TIR,SF}})$ show that radio-excess hosts are more compact than controls (Gini $d = +0.22$; concentration $d = +0.18$) and smoother (asymmetry $d = -0.14$). The Galaxy Zoo Hubble spiral deficit ($d = -0.434$; $p = 1.6 \times 10^{-10}$) is the largest single-feature effect in either system. Among flat-spectrum sources ($N_{\text{rx}} = 139$), the asymmetry suppression is extreme: $d_{\text{Asym}} = -0.867$ (95% CI $[-1.118, -0.637]$; $p = 8 \times 10^{-13}$), robust to spectral-index threshold cuts up to $2\sigma_\alpha$. A parametric Sérsic validation yields $\Delta R^2 = 0.004$, confirming the signal is specific to non-parametric structure.

Two independent systems converge on the same physical picture: radio-excess galaxies are structurally regular, non-disk-dominated, and compact. A single smooth/featured classification from Euclid, Rubin LSST, or Roman will deliver the dominant improvement to radio-SFR calibration in the SKA era.

Keywords: galaxies: star formation — galaxies: structure — radio continuum: galaxies — galaxies: evolution — galaxies: active

1. INTRODUCTION

Radio continuum emission is one of astronomy’s most reliable star-formation tracers because it penetrates dust. Synchrotron radiation from cosmic-ray (CR) electrons accelerated in supernova remnants, combined with thermal free-free emission from ionised gas, traces recent massive star formation where optical photometry fails (Condon 1992). The empirical infrared-radio correlation (IRRC), linking radio luminosity to total infrared luminosity and

thence to SFR, has underpinned decades of cosmic star-formation census work (Helou et al. 1985; Yun et al. 2001; Bell 2003; Murphy et al. 2011; Madau & Dickinson 2014).

The IRRC is not a perfect one-parameter mapping. Its $\sigma \sim 0.2\text{--}0.3$ dex scatter encodes genuine physics: the synchrotron yield per unit SFR depends on (i) the calorimetric fraction, which rises with gas surface density and infrared luminosity (Völk 1989; Lacki et al. 2010); (ii) the magnetic field strength, setting the synchrotron cooling timescale (Schleicher & Beck 2013); (iii) inverse-Compton losses off the cosmic microwave background at high redshift (Lacki et al. 2010; Schober et al. 2016); and (iv) the free-free fraction, varying with frequency and ionisation state (Condon 1992). Every one of these processes is regulated by structure: a compact, high-surface-density system radiates more synchrotron per unit SFR than a diffuse irregular of the same mass.

Observational evidence for this structural dependence has grown. Magnelli et al. (2015) found a trend of q_{IR} with stellar mass. Delhaize et al. (2017) detected redshift evolution of q_{IR} , and Delvecchio et al. (2021) showed that q_{IR} depends jointly on mass and redshift consistent with inverse-Compton losses dominating at high z . These results confirm the IRRC is not universal, but the structural degrees of freedom driving the scatter have never been isolated in a controlled, cross-validated framework separating them from the global scalars they correlate with. A second practical complication is AGN contamination: radio-excess sources (Del Moro et al. 2013) must be identified structurally to avoid biasing radio-SFR calibrations for next-generation surveys.

Non-parametric morphology offers a principled route to both problems. The Gini coefficient (Abraham et al. 2003), the concentration-asymmetry system (Conselice 2003), and the Gini- M_{20} framework (Lotz et al. 2004) encode compactness, nuclear dominance, disturbance, and clumpiness without parametric assumptions. Concentration is a direct proxy for the gas surface densities regulating CR calorimetric efficiency (Thompson et al. 2006; Lacki et al. 2010); asymmetry traces dynamical disturbance long proposed as an AGN trigger (Lotz et al. 2008; Ellison et al. 2013); Gini captures nuclear light concentration. Crucially, these statistics are computed directly from the pixel flux distribution without assuming any functional form for the light profile, making them sensitive to substructure — nuclear point sources, asymmetric tidal features, star-forming clumps — that a smooth parametric model cannot capture.

The natural alternative is parametric morphology: fitting a Sérsic profile (Sérsic 1968) with GALFIT (Peng et al. 2002) to extract the Sérsic index n and effective radius R_e . This approach is widely used in morphological studies and delivers physically interpretable quantities; n encodes the degree of central concentration and R_e sets the physical scale. However, Sérsic fitting describes only the azimuthally averaged, smooth component of the light profile and is by construction insensitive to the rotational asymmetry, nuclear excess, and fine-grained radial structure that non-parametric statistics capture. Whether this distinction matters quantitatively for radio-SFR calibration is not established. We test both approaches on the same galaxy sample and find the contrast decisive: non-parametric morphology delivers $32\times$ more predictive information than Sérsic parameters from the same imaging data.

Citizen-science classifications from Galaxy Zoo (Lintott et al. 2008; Willett et al. 2013) provide an independent, scalable route to the same structural information. If the morphological signal is real, it should appear in volunteer vote fractions derived from a completely different measurement process. Makechemu et al. (2026) establish the complementary bound: six regression architectures spanning linear to deep non-linear models all saturate at $R^2 \lesssim 0.45$ from radio luminosity alone, confirming the dominant scatter is physical rather than methodological. Morphological information does not merely improve calibration but fills a gap that no modelling flexibility can close.

The COSMOS field (Scoville et al. 2007) provides the ideal testbed. The VLA-COSMOS 3 GHz Large Project (Smolčić et al. 2017) delivers $\sim 2.3 \mu\text{Jy beam}^{-1}$ sensitivity at $0''.75$ resolution over $\sim 2 \text{ deg}^2$, with multiwavelength AGN classification (Delvecchio et al. 2017; Novak et al. 2017). The Zurich/Tasca HST/ACS non-parametric catalogue (Tasca et al. 2009) and the Galaxy Zoo Hubble citizen-science catalogue (Willett et al. 2017) both cover this field at sufficient depth for a controlled comparison.

We pursue four questions: (1) what is the incremental predictive hierarchy of physical information beyond radio luminosity? (2) which morphological parameters carry the strongest independent signal for radio-excess behaviour? (3) does a citizen-science classification system recover the same structural signal as precision HST measurements? (4) does parametric Sérsic morphology deliver comparable information to non-parametric statistics from the same imaging data?

2. DATA AND SAMPLE

2.1. VLA-COSMOS 3 GHz Catalogue

We draw our parent sample from the VLA-COSMOS 3 GHz Large Project catalogue (Smolčić et al. 2017), extracting sky positions, photometric redshifts from COSMOS2015 (Laigle et al. 2016), stellar masses and SF-component infrared luminosities ($\log L_{\text{TIR,SF}}$) from SED fitting (Delvecchio et al. 2017), rest-frame radio luminosities at 1.4 and 3 GHz, multiwavelength AGN flags, and the radio-excess flag. After quality cuts, $N = 7729$ sources remain.

2.2. Zurich/Tasca Morphology Catalogue

The COSMOS morphology catalogue (Tasca et al. 2009), constructed from HST/ACS *I*-band imaging at $0''.03 \text{ pix}^{-1}$ resolution, provides the Gini coefficient (Abraham et al. 2003), concentration and asymmetry (Conselice 2003), half-light radius R_{HALF} , axial ratio q , and an SVM-based morphological class. We convert R_{HALF} to physical half-light radius R_e under the Planck 2018 cosmology (Planck Collaboration et al. 2020), define a stellar surface-density proxy $\log \Sigma_\star \equiv \log(M_\star/M_\odot) - \log(2\pi R_e^2)$, and an inclination proxy $i_{\text{proxy}} \equiv \arccos(q)$.

2.3. Galaxy Zoo Hubble

The Galaxy Zoo Hubble catalogue (Willett et al. 2017) provides debiased and weighted volunteer vote fractions from non-expert visual classification of HST images. We retain 84,954 COSMOS sources and sky-match to the VLA catalogue within $0''.7$, yielding $N_{\text{match}} = 3696$ (median separation 0.14 arcsec). The primary predictor is the debiased featured-or-disk fraction ($t01$), which corrects for the known bias that volunteers classify high-redshift galaxies as smooth at a rate exceeding their true structural smoothness (Bamford et al. 2009). Additional vote fractions used in the ladder are: odd features ($t06$), merger ($t08$), disturbed ($t08$), spiral ($t04$), bulge dominance ($t05$), edge-on probability ($t02$), and clumpy fraction ($t12$).

2.4. Spectral Index, Compactness, and Sample Construction

A two-band spectral index is computed as $\alpha_{\text{radio}} = (\log L_{3 \text{ GHz}} - \log L_{1.4 \text{ GHz}}) / \log_{10}(3.0/1.4)$. Because both luminosities were derived under a fixed $\alpha = -0.7$ (Delvecchio et al. 2017), α_{radio} has limited dynamic range (median $= -0.700$) and should be regarded as a qualitative indicator of emission character. We use it to define flat-spectrum ($\alpha_{\text{radio}} \geq -0.5$; $N = 278$; 5.7%) and steep-spectrum ($\alpha_{\text{radio}} < -0.5$; $N = 4565$; 94.3%) sub-samples. An independent geometry-based compactness proxy, the radio-to-optical size ratio $\text{ROR} \equiv \log_{10}[\theta_{\text{beam}} / (2\sqrt{2 \ln 2} \cdot R_{\text{HALF},\nu})]$ with $\theta_{\text{beam}} = 0''.75$, agrees with the spectral-index classification for 52.3% of sources, confirming the two proxies measure different physical dimensions.

Sky-matching VLA and Zurich catalogues within $0''.7$ yields $N = 4844$ sources; enforcing complete coverage in all ladder variables gives the *fixed matched sample* of $N = 4843$ galaxies used throughout (Table 1). All ladder experiments use this fixed sample, so performance differences between rungs cannot reflect changes in sample composition.

3. METHODS

3.1. Information Ladder

Our regression target is $\log L_{\text{TIR,SF}}$. We define cumulative feature sets in which each rung retains all features from lower rungs. For the Zurich analysis the eight rungs are: (1) $L_{1.4 \text{ GHz}}$ only; (2) $+L_{3 \text{ GHz}}$; (3) $+\log(M_\star/M_\odot)$; (4) $+R_e$; (5) $+\log \Sigma_\star$; (6) $+$ Gini, concentration, asymmetry; (7) $+i_{\text{proxy}}$; (8) $+\text{CLASS_SVM}$. The Galaxy Zoo Hubble ladder adds each of its eight vote fractions sequentially from the same radio-mass baseline.

The primary inference model is a multilayer perceptron (MLP; two hidden layers of 64 and 32 neurons, ReLU activations, Adam optimisation, early stopping on a 10% validation split). All results use 5-fold cross-validation with seed = 42. A *redshift-sensitivity check* is applied to both ladders: redshift is inserted explicitly after stellar mass to quantify how much of the morphological ΔR^2 reflects redshift-correlated trends rather than genuine structural information. The resulting retention fraction — the ratio of ΔR^2 with z controlled to ΔR^2 without — measures the degree to which the morphological gain survives independent of cosmic evolution.

3.2. Radio-Excess Classification and Matched-Sample Tests

We evaluate logistic-regression ROC-AUC (5-fold stratified) for individual and joint morphology parameters, and nested models extending $\{\log(M_\star/M_\odot), z, \log L_{\text{TIR,SF}}\}$. BIC from maximum-likelihood fits provides penalised model selection.

Table 1. Fixed Matched Sample Properties

Quantity	Value
Sample sizes	
VLA working sample (quality cuts)	7729
Zurich catalogue (coord. cleaning)	237,912
Zurich matched within 0.''7	4844
Fixed ladder sample (N)	4843
GZ Hubble matched within 0.''7	3696
Medians (fixed sample)	
Redshift z	0.91
$\log(M_*/M_\odot)$ ($M_\odot$)	10.87
$\log L_{1.4\text{ GHz}}$ (W Hz^{-1})	23.20
$\log L_{3\text{ GHz}}$ (W Hz^{-1})	22.97
R_e (kpc)	2.51
Gini / Concentration / Asymmetry	0.540 / 0.342 / 0.086
Activity classes (binary radio-excess flag)	
Non-radio-excess (N_{non})	3840
Radio-excess (N_{rx})	1003

The central test is a **unique one-to-one nearest-neighbour matched-sample comparison**: each radio-excess source is matched to its closest non-radio-excess counterpart in standardised ($\log(M_*/M_\odot)$, z , $\log L_{\text{TIR,SF}}$) space with no replacement, producing $N_{\text{rx}} = N_{\text{non}} = 1003$ independent pairs (Zurich) or 697 pairs (Galaxy Zoo Hubble). Morphological differences are quantified by Cohen’s d with 95% bootstrap confidence intervals (5000 resamples for Zurich; 2000 for Galaxy Zoo Hubble).

4. RESULTS

4.1. Two Systems, One Signal

Table 2 presents the full quantitative results. Figure 1 shows the two information ladders side by side.

Zurich non-parametric morphology. Starting from radio luminosity alone, the MLP achieves $R^2 = 0.465$. Adding a second radio band, stellar mass, physical size, and surface density collectively recover modest additional variance. At rung 6, adding Gini, concentration, and asymmetry simultaneously produces:

$$\Delta R^2_{\text{morph}} = +0.127 \quad (\text{MLP}), \quad (1)$$

compared to +0.120 for the linear model. This single step raises cumulative R^2 from 0.495 to 0.621 and reduces per-source scatter from 0.424 dex to 0.367 dex, a 13% RMSE improvement from three numbers alone. The MLP gain is $4.3\times$ the combined contribution of all other scalar variables. Subsequent rungs (inclination, $\Delta R^2 = -0.010$; morphological class, $\Delta R^2 = +0.003$) add nothing once the continuous metrics are included.

Redshift-sensitivity check. When redshift is inserted after stellar mass (Figure 2), z alone contributes $\Delta R^2 = +0.108$, reducing $\Delta R^2_{\text{morph}}$ to +0.038 (retention fraction 0.30). This residual remains the single largest per-variable gain from any observable, confirming morphology encodes genuine structural information beyond redshift-correlated evolution.

Galaxy Zoo Hubble. The debiased featured/disk vote fraction alone delivers $\Delta R^2 = +0.156$ (Figure 1), the largest single-variable gain across both systems. All subsequent Galaxy Zoo Hubble vote fractions contribute negligibly once

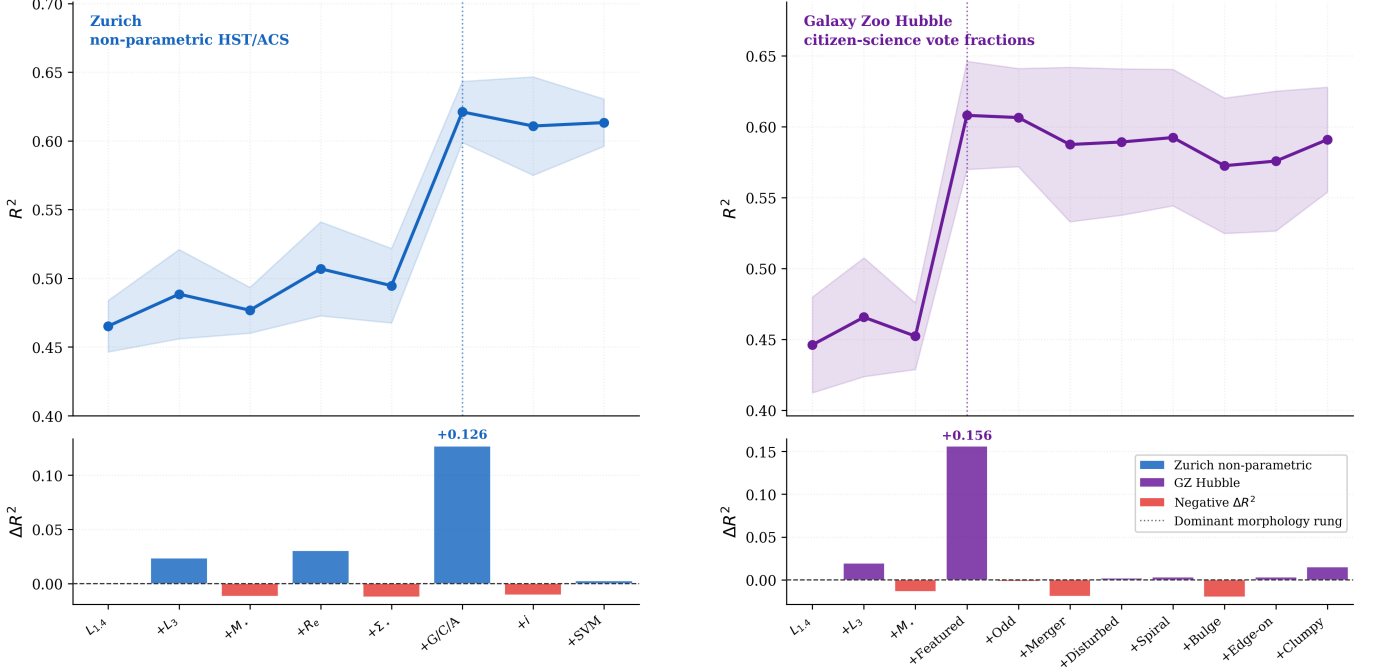


Figure 1. Information ladders for the two morphology systems (5-fold CV MLP; seed = 42; target: $\log L_{\text{TIR,SF}}$). Left column: Zurich non-parametric HST/ACS morphology ($N = 4843$). Right column: Galaxy Zoo Hubble citizen-science vote fractions ($N = 3696$). Top panels show cumulative R^2 with $\pm 1\sigma$ shading; bottom panels show incremental ΔR^2 bars (blue/purple for positive gains, red for negative). The dotted vertical line marks the dominant morphology rung in each system. Rungs from left to right: (Zurich) $L_{1.4}$, $+L_3$, $+M_*$, $+R_e$, $+z$, $+G/C/A$, $+i_{\text{proxy}}$, $+CLASSSVM$; (GZ Hubble) $L_{1.4}$, $+L_3$, $+M_*$, $+featured/disk$, $+odd$, $+merger$, $+disturbed$, $+spiral$, $+bulge$, $+edge-on$, $+clumpy$. The featured/disk rung (right column, 4th bar) yields $\Delta R^2 = +0.156$ from a single volunteer vote fraction, matching the Zurich morphology step ($\Delta R^2 = +0.127$, left column, 6th bar).

this rung is included. One coarse volunteer classification recovers more predictive information than the full Zurich non-parametric suite from a completely independent instrument and measurement process.

Galaxy Zoo Hubble redshift-sensitivity check. Applying the same redshift-sensitivity check to the GZ Hubble ladder yields a qualitatively different and physically informative result. When redshift is inserted after stellar mass, z alone contributes $\Delta R^2 = +0.096$, reducing the featured/disk ΔR^2 from $+0.156$ to $+0.078$ (retention fraction 0.50; Figure 3). This is substantially higher than the Zurich retention fraction of 0.30. The residual $\Delta R^2 = 0.078$ also exceeds the Zurich residual of 0.038 in absolute terms, making the redshift-controlled GZ Hubble gain the largest per-variable structural improvement in either system.

The higher GZ Hubble retention fraction reflects the debiasing correction (Bamford et al. 2009), which removes the dominant redshift trend from volunteer vote fractions before the ladder is run; the pixel-level Zurich metrics carry a larger unremoved redshift component from both genuine structural evolution and measurement degradation with angular resolution. The physical interpretation and its implications for the convergence argument are developed in Section 5.

Quantitative convergence. The combined AUC from all Galaxy Zoo Hubble features is 0.697 ± 0.009 ; from all Zurich non-parametric features, 0.693 ± 0.009 . Two systems sharing no instruments, algorithms, or classifiers agree to within the measurement uncertainty of both. This convergence is the primary evidence that the signal is a real physical property of the galaxies.

4.2. Matched-Sample Morphological Tests

Unique nearest-neighbour matching produces 1003 radio-excess and 1003 non-radio-excess control pairs (Zurich), and 697 pairs (Galaxy Zoo Hubble). Figure 4 presents the Cohen’s d results across both systems.

Table 2. Information Ladder and AUC: Zurich and Galaxy Zoo Hubble

Rung / Feature	System	N	ΔR^2 (MLP)	R^2 (MLP)	AUC
Zurich non-parametric (5-fold CV; seed = 42)					
Baseline: $L_{1.4\text{ GHz}}$	Zurich	4843	...	0.465	...
+ $L_{3\text{ GHz}}$	Zurich	4843	+0.023	0.489	...
+ $\log(M_\star/M_\odot)$	Zurich	4843	-0.012 ^a	0.477	...
+ R_e	Zurich	4843	+0.030	0.507	...
+ $\log \Sigma_\star$	Zurich	4843	-0.012 ^a	0.495	...
+Morphology (G, C, A)	Zurich	4843	+0.127	0.621	...
+ i_{proxy}	Zurich	4843	-0.010 ^a	0.611	...
+ CLASS_SVM	Zurich	4843	+0.003	0.613	...
Redshift-sensitivity check (z after $\log(M_\star/M_\odot)$; MLP)					
+ z (rung 3b)	Zurich	4843	+0.108	0.603	...
+Morphology (with z controlled)	Zurich	4843	+0.038	0.652	...
Galaxy Zoo Hubble (5-fold CV; seed = 42)					
Baseline: $L_{1.4\text{ GHz}}$	GZ Hub.	3696	...	0.446	...
+ $L_{3\text{ GHz}}$	GZ Hub.	3696	+0.020	0.466	...
+ $\log(M_\star/M_\odot)$	GZ Hub.	3696	-0.013 ^a	0.452	...
+GZ: Featured/Disk	GZ Hub.	3696	+0.156	0.608	...
All subsequent GZ features	GZ Hub.	3696	$\leq \pm 0.020^a$	0.608	...
AUC (5-fold stratified logistic regression)					
Asymmetry	Zurich	4843	...	...	0.658 ± 0.016
All Zurich morphology (5 params.)	Zurich	4843	...	...	0.693 ± 0.009
Spiral fraction	GZ Hub.	3696	...	...	0.660 ± 0.015
All GZ Hubble features	GZ Hub.	3696	...	...	0.697 ± 0.009
Nested logistic model (Zurich; $N = 4843$)					
Base: $\log(M_\star/M_\odot) + z + \log L_{\text{TIR,SF}}$	Zurich	4843	...	...	0.865 ± 0.015
Full: base + Gini, Conc., Asym.	Zurich	4843	...	...	0.865 ± 0.015

NOTE—^a Negative ΔR^2 values reflect stochastic early-stopping convergence in the Adam optimiser, not genuine information loss. Bold rungs are the only ones from which structural conclusions are drawn. Morphology ΔR^2 exceeds the summed pre-morphology scalar contributions by $4.3\times$ (Zurich MLP) and $15.3\times$ (Zurich linear). Retention fraction with z controlled: Zurich $0.038/0.127 = 0.30$; GZ Hubble $0.078/0.156 = 0.50$. The nested model BIC increases by +25 upon adding morphology, indicating population mixing absorbs the full-sample effect; the matched-sample test in Section 4.2 isolates the residual structural signal.

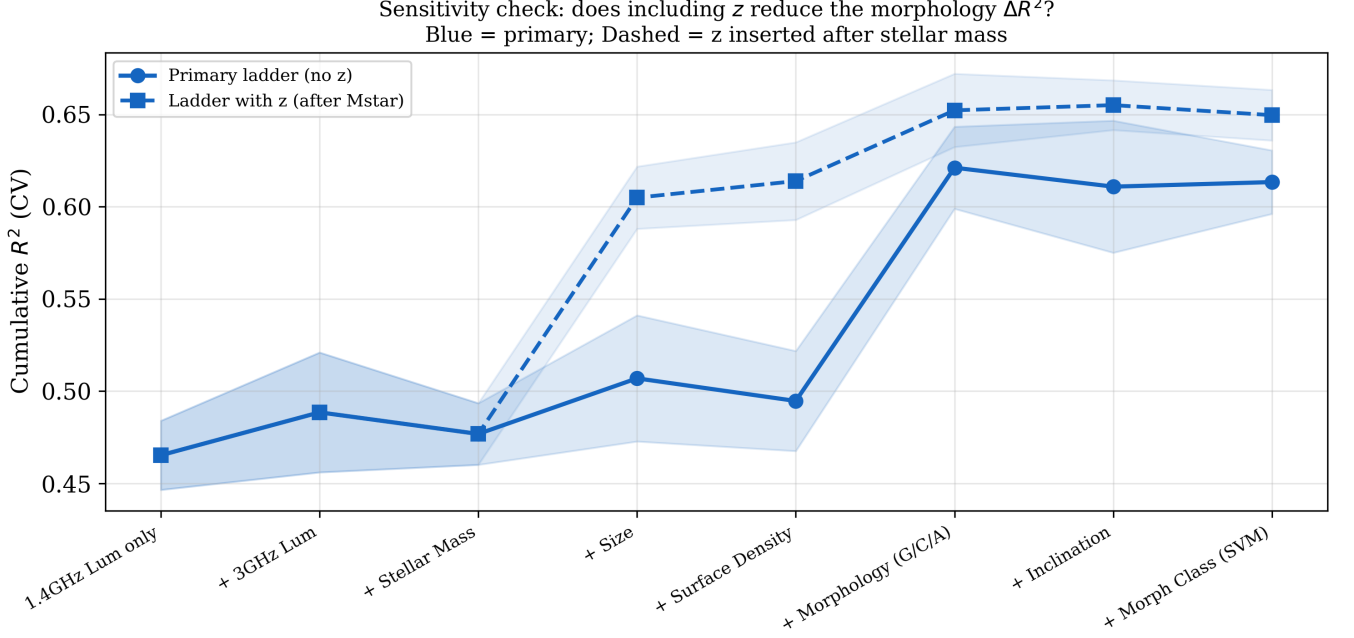


Figure 2. Redshift-sensitivity check on the Zurich information ladder (MLP; $N = 4843$). Solid circles: primary ladder without explicit redshift. Open squares: ladder with redshift inserted after stellar mass. Shaded bands show $\pm 1\sigma$ from 5-fold cross-validation. Inserting z captures $\Delta R^2 = +0.108$ before the morphology rung, reducing $\Delta R^2_{\text{morph}}$ from 0.127 to 0.038 (retention fraction 0.30). The surviving morphological gain remains the single largest per-variable improvement available from any observable.

Zurich full sample. Radio-excess hosts are more compact than controls: Gini $d = +0.22$ (95% CI $[+0.13, +0.31]$; $p = 4.6 \times 10^{-6}$); concentration $d = +0.18$ ($[+0.10, +0.27]$; $p = 6.2 \times 10^{-4}$). Asymmetry shows a significant negative offset: $d = -0.14$ ($[-0.23, -0.05]$; $p = 3.6 \times 10^{-4}$). Radio-excess hosts are smoother and more centrally concentrated at fixed mass, redshift, and SFR.

Galaxy Zoo Hubble full sample. Radio-excess hosts show a significantly lower featured/disk fraction ($d = -0.131$; $p = 1.8 \times 10^{-3}$) and the largest single-feature matched-sample effect in either system: a dramatically reduced spiral fraction ($d = -0.434$; 95% CI $[-0.538, -0.332]$; $p = 1.6 \times 10^{-10}$). The clumpy fraction is significantly elevated ($d = +0.236$; $p = 7.9 \times 10^{-6}$), consistent with a bright compact nucleus. These are volunteer-language translations of the same compactness and smoothness signal identified in the Zurich analysis.

4.3. The Sign Flip: Spectral Stratification Reveals Two Populations

Splitting the Zurich matched sample by spectral index reveals a physically decisive differentiation (Table 3).

Among steep-spectrum sources ($\alpha_{\text{radio}} < -0.5$; $N_{\text{rx}} = 864$), Gini ($d = +0.19$) and concentration ($d = +0.16$) retain their positive offsets; asymmetry shows a mild negative offset ($d = -0.12$; $p = 0.011$), consistent with a slight preference for smooth hosts in AGN populations after mass-and-SFR control (Cisternas et al. 2011; Mechtley et al. 2016).

Among flat-spectrum sources ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$), the signal changes qualitatively and dramatically. Gini ($d = +0.28$) and concentration ($d = +0.35$) retain their positive offsets, while asymmetry shows a suppression of extraordinary magnitude:

$$d_{\text{Asym}} = -0.867, \quad 95\% \text{ CI } [-1.118, -0.637], \quad p = 8.0 \times 10^{-13}. \quad (2)$$

The median asymmetry of radio-excess flat-spectrum hosts ($\langle A \rangle_{\text{rx}} = 0.059$) is barely half that of their individually matched controls ($\langle A \rangle_{\text{non}} = 0.110$). A Cohen’s d of -0.87 exceeds the conventional large-effect threshold ($|d| > 0.8$); Figure 5 shows it is visible across every projection of the morphological parameter space.

The radio-to-optical size ratio provides independent corroboration: across the full sample, radio-excess sources are more compact relative to their optical size ($d = +0.41$; $p = 1.3 \times 10^{-25}$).

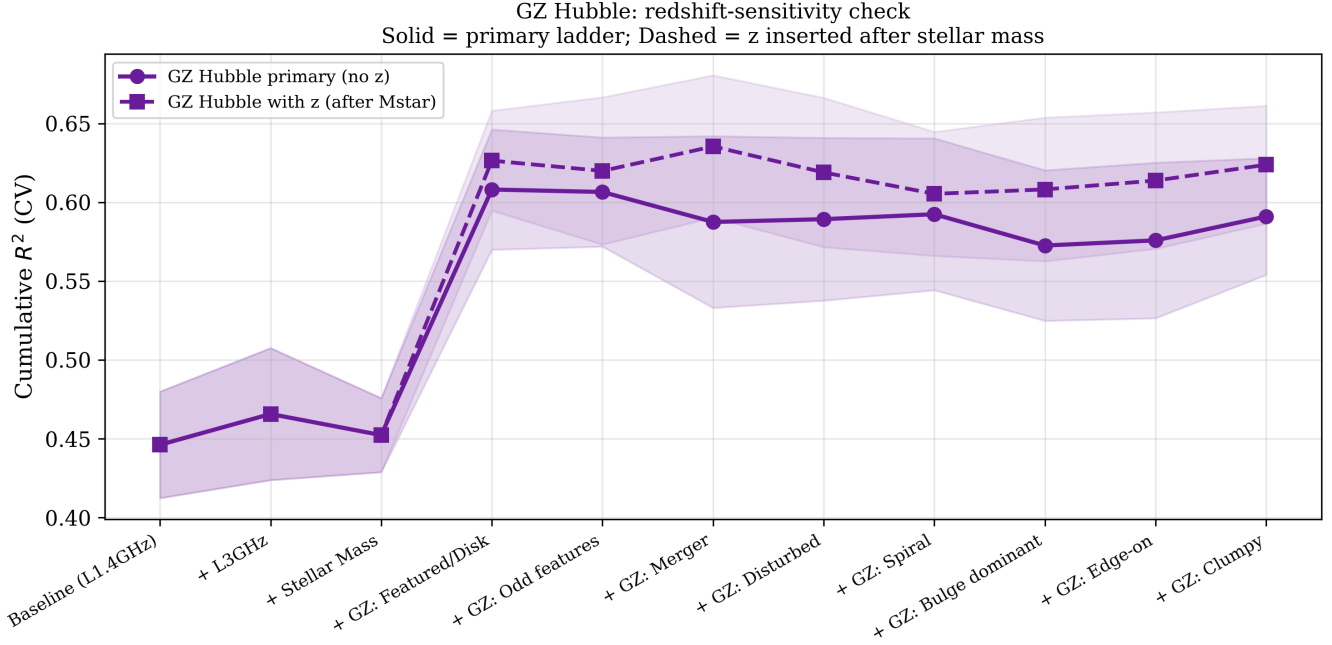


Figure 3. Redshift-sensitivity check on the Galaxy Zoo Hubble information ladder (MLP; $N = 3696$). Solid circles: primary ladder without explicit redshift. Dashed squares: ladder with redshift inserted after stellar mass. Shaded bands show $\pm 1\sigma$ from 5-fold cross-validation. Inserting z captures $\Delta R^2 = +0.096$ before the featured/disk rung, reducing its ΔR^2 from $+0.156$ to $+0.078$ (retention fraction 0.50). This retention fraction substantially exceeds the Zurich value of 0.30, reflecting the fact that the debiased vote fraction has had the dominant redshift trend in volunteer behaviour removed by construction. The residual $\Delta R^2 = 0.078$ also exceeds the Zurich residual of 0.038 in absolute terms, making it the largest per-variable structural gain in either system after redshift control.

4.4. Robustness of the Flat-Spectrum Signal

Both luminosities were derived under a fixed $\alpha = -0.7$, so the flat-spectrum sub-sample partly selects on scatter around that assumed value. Table 4 tests this systematically. The asymmetry suppression is unchanged at $\geq 1\sigma_\alpha$ ($d = -0.856$; $p = 1.7 \times 10^{-12}$; $N_{\text{rx}} = 138$), remains highly significant at $\geq 1.5\sigma_\alpha$ ($d = -0.854$; $p = 1.2 \times 10^{-7}$; $N_{\text{rx}} = 89$), and persists at $\geq 2\sigma_\alpha$ ($d = -0.573$; $p = 0.003$; $N_{\text{rx}} = 56$), despite removing more than half the original sub-sample. A continuous test within the flat-spectrum radio-excess sub-sample finds no significant trend of asymmetry residual with spectral-index deviation (Spearman $\rho = +0.033$; $p = 0.70$): the suppression is a uniform population-level property, not a boundary artefact.

Figure 6 shows the full bootstrap sampling distribution for d_{Asym} ; every one of the 5000 resamples falls below zero.

4.5. Redshift Evolution and Completeness

Both morphological cross-match completeness and morphology-only AUC decline monotonically across all four measured redshift bins: AUC falls from 0.799 ± 0.046 at $z < 0.5$ to 0.575 ± 0.038 at $1.5 < z < 2.0$, while completeness drops from $\sim 85\%$ to $\sim 38\%$ over the same interval. In the broad-bin analysis, AUC declines from 0.748 ± 0.017 at $0 < z < 1$ to 0.622 ± 0.020 at $1 < z < 2$ ($\Delta\text{AUC} = -0.127$; $p = 1.2 \times 10^{-5}$; Figure 7). The completeness trend alone is sufficient to explain the AUC decline without invoking structural evolution; a genuine evolutionary contribution cannot be excluded but would require deeper space-based imaging to resolve.

5. DISCUSSION

5.1. Convergence Is the Key Result

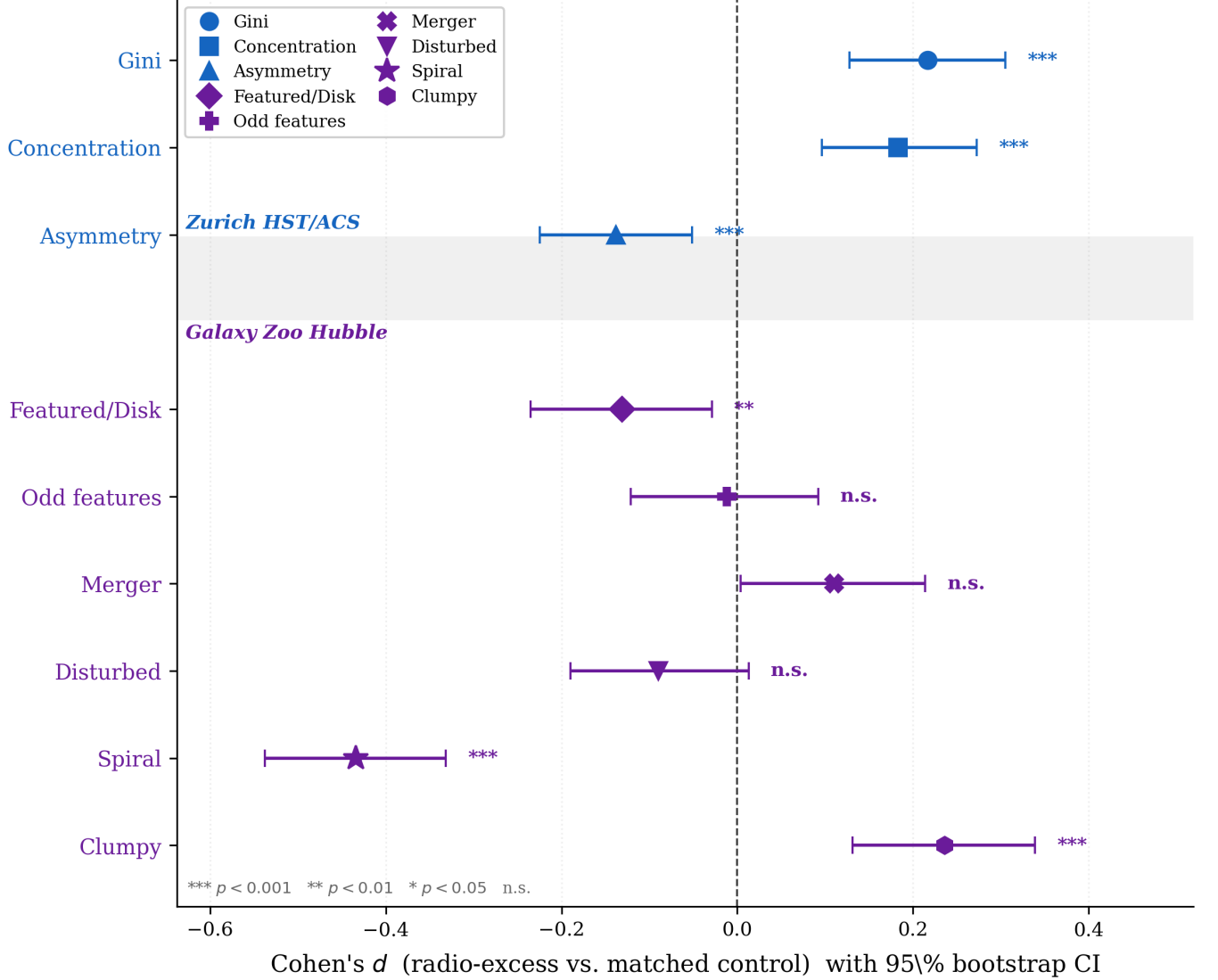


Figure 4. Matched-sample Cohen’s d (radio-excess vs. matched controls) from two independent morphology systems. Error bars are 95% bootstrap confidence intervals. The shaded band separates the two system blocks. Top block (blue): Zurich non-parametric ($N_{\text{rx}} = 1003$ unique pairs; 5000 resamples); rows from top to bottom are Gini (circle), Concentration (square), Asymmetry (triangle). Bottom block (purple): Galaxy Zoo Hubble ($N_{\text{rx}} = 697$ unique pairs; 2000 resamples); rows from top to bottom are Featured/Disk (diamond), Odd features (plus), Merger (cross), Disturbed (down-triangle), Spiral (star), Clumpy (hexagon). The vertical dashed line marks $d = 0$. All pairs matched 1:1 in $(\log M_*, z, \log L_{\text{TIR,SF}})$; no replacement.

AUC values of 0.693 and 0.697 from instruments, algorithms, and classifiers that share nothing except the galaxies being observed do not coincide by chance. The probability of two independent measurement systems producing this agreement on a spurious signal is negligible. The structural signal is real, and its physical origin must be sought in the galaxies themselves rather than in any measurement artefact.

The redshift-sensitivity check reinforces this conclusion in a second dimension. The Zurich non-parametric metrics retain 30% of their ΔR^2 after explicit redshift control; the Galaxy Zoo Hubble debiased featured/disk fraction retains 50%, and in absolute terms its residual ($\Delta R^2 = 0.078$) is more than twice the Zurich residual ($\Delta R^2 = 0.038$). The higher retention fraction is physically expected: the debiasing correction (Bamford et al. 2009) has already removed the dominant redshift trend from the vote fractions by construction, whereas the pixel-level asymmetry and concentration metrics carry an unremoved component from both genuine structural evolution and measurement degradation with angular resolution. Far from being a weakness, the higher GZ Hubble retention fraction demonstrates that citizen-

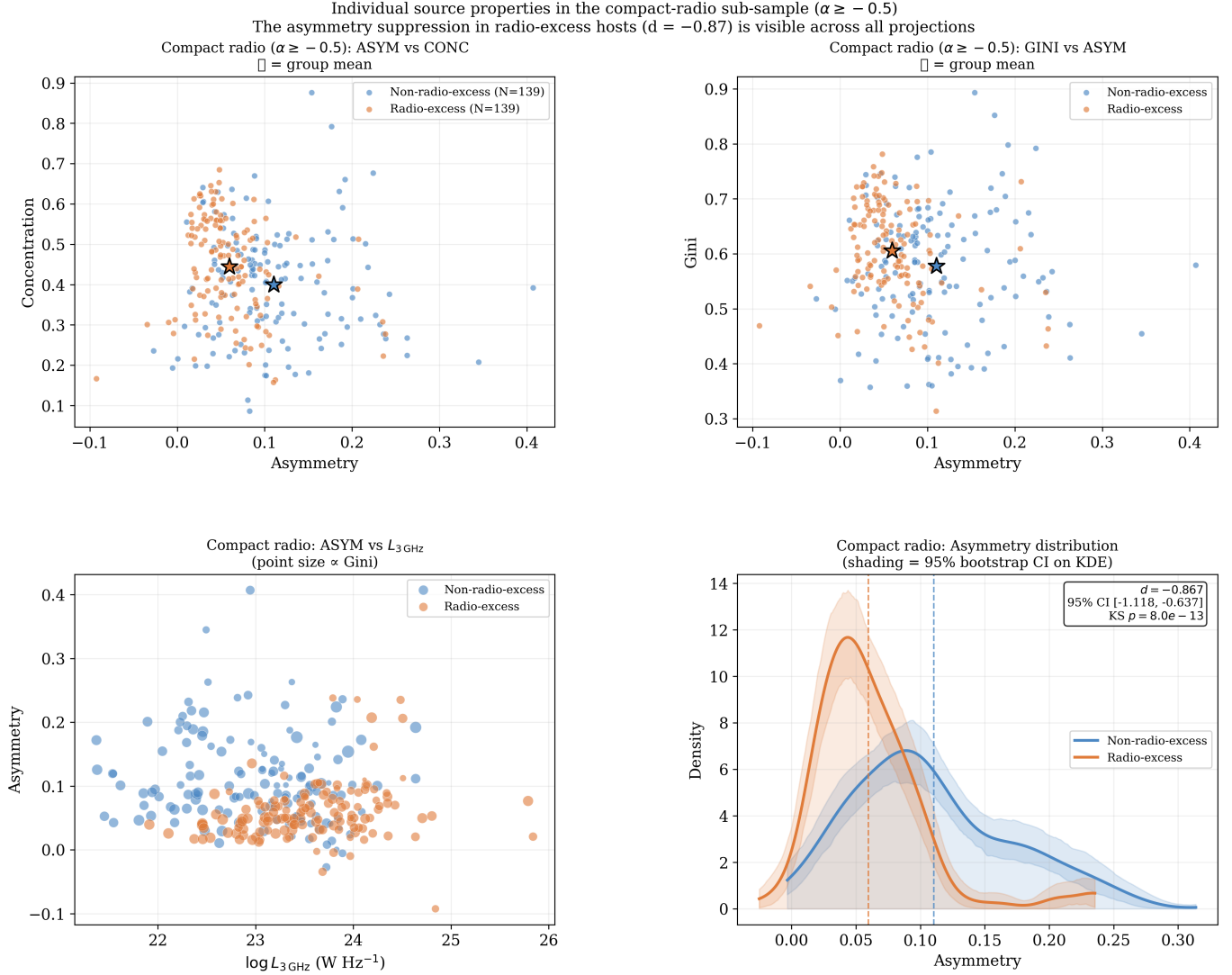


Figure 5. Individual source properties in the flat-spectrum sub-sample ($\alpha_{\text{radio}} \geq -0.5$; $N = 278$). Orange points: radio-excess ($N_{\text{rx}} = 139$). Blue points: non-radio-excess ($N = 139$). Stars mark group means. *Top left:* asymmetry vs. concentration; radio-excess sources cluster at low asymmetry across the full concentration range. *Top right:* Gini vs. asymmetry; asymmetry is the dominant separation axis. *Bottom left:* asymmetry vs. $\log L_{3\text{ GHz}}$ (point size $\propto$ Gini); no luminosity dependence of the asymmetry offset is apparent. *Bottom right:* 1D asymmetry KDEs with 95% bootstrap CI shading; Cohen’s $d = -0.867$ ($p = 8 \times 10^{-13}$; $N_{\text{rx}} = 139$ unique matched pairs; 5000 resamples). The suppression is pronounced and consistent across all projections.

science debiasing acts as a partial redshift pre-correction, leaving a morphological residual that is more genuinely structural than the raw non-parametric measurements.

The convergence also defines the practical minimum measurement required. The Galaxy Zoo Hubble featured/disk fraction achieves $\Delta R^2 = 0.156$ — or $\Delta R^2 = 0.078$ after redshift control — from a single yes/no classification. This fraction summarises, in compressed volunteer-language form, the same information that Gini, concentration, and asymmetry encode continuously. A galaxy with high Gini and high concentration is likely to be voted smooth; one with high asymmetry is likely to be voted featured. The coarse classification is sufficient to capture the dominant predictive variance, and it does so with a larger redshift-controlled residual than three precisely measured pixel statistics.

5.2. Two Physical Channels in the Morphology Step

The $\Delta R^2 = 0.127$ from Gini, concentration, and asymmetry arises through two distinct mechanisms. The first is *calorimetric*. Gini and concentration are proxies for gas surface density and magnetic-field strength: compact systems

Table 3. Bootstrap Cohen’s d : Zurich and Galaxy Zoo Hubble

Feature / Sub-sample	d	95% CI	KS p
Zurich non-parametric (1003 unique pairs; 5000 resamples)			
<i>Full sample</i> ($N_{\text{rx}} = 1003$)			
Gini	+0.22	[+0.13, +0.31]	4.6×10^{-6}
Concentration	+0.18	[+0.10, +0.27]	6.2×10^{-4}
Asymmetry	−0.14	[−0.23, −0.05]	3.6×10^{-4}
<i>Flat-spectrum</i> ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$)			
Gini	+0.28	[+0.05, +0.53]	1.5×10^{-2}
Concentration	+0.35	[+0.11, +0.60]	1.0×10^{-2}
Asymmetry	−0.87	[−1.12, −0.64]	8.0×10^{-13}
<i>Steep-spectrum</i> ($\alpha_{\text{radio}} < -0.5$; $N_{\text{rx}} = 864$)			
Gini	+0.19	[+0.10, +0.29]	3.1×10^{-4}
Concentration	+0.16	[+0.07, +0.26]	1.2×10^{-3}
Asymmetry	−0.12	[−0.22, −0.03]	1.1×10^{-2}
Galaxy Zoo Hubble (697 unique pairs; 2000 resamples; full sample)			
Featured/Disk (debiased)	−0.13	[−0.24, −0.03]	1.8×10^{-3}
Odd features	−0.01	[−0.12, +0.09]	8.0×10^{-1}
Merger	+0.11	[+0.00, +0.21]	2.3×10^{-1}
Spiral	−0.43	[−0.54, −0.33]	1.6×10^{-10}
Clumpy	+0.24	[+0.13, +0.34]	7.9×10^{-6}

NOTE—All matches are unique one-to-one in $(\log(M_{\star}/M_{\odot}), z, \log L_{\text{TIR,SF}})$ with no replacement.

Table 4. Robustness of Flat-Spectrum Asymmetry Suppression ($\sigma_{\alpha} = 0.20$)

Threshold	N_{rx}	d	KS p	Status
Unrestricted	139	−0.867	8.0×10^{-13}	Reference
$\geq 0.5\sigma_{\alpha}$	139	−0.867	8.0×10^{-13}	Unchanged
$\geq 1.0\sigma_{\alpha}$	138	−0.856	1.7×10^{-12}	PERSISTS
$\geq 1.5\sigma_{\alpha}$	89	−0.854	1.2×10^{-7}	PERSISTS
$\geq 2.0\sigma_{\alpha}$	56	−0.573	2.96×10^{-3}	persists

NOTE—Continuous test within the flat-spectrum radio-excess sub-sample: Spearman $\rho = +0.033$, $p = 0.70$.

approach the calorimetric limit, with CR electrons losing most of their energy to synchrotron radiation before escaping (Völk 1989; Lacki et al. 2010). Partial correlations of Gini ($r = +0.048$) and concentration ($r = +0.053$) with α_{radio} after removing mass and redshift trends (both $p < 0.001$) confirm this channel is active. A global scalar like $\log \Sigma_{\star}$ cannot capture this: a compact bulge-dominated elliptical, a nuclear ring disk, and a disturbed irregular can share the same $\log \Sigma_{\star}$ while differing radically in synchrotron yield. The MLP exploits these within- $\log \Sigma_{\star}$ variations; linear regression cannot.

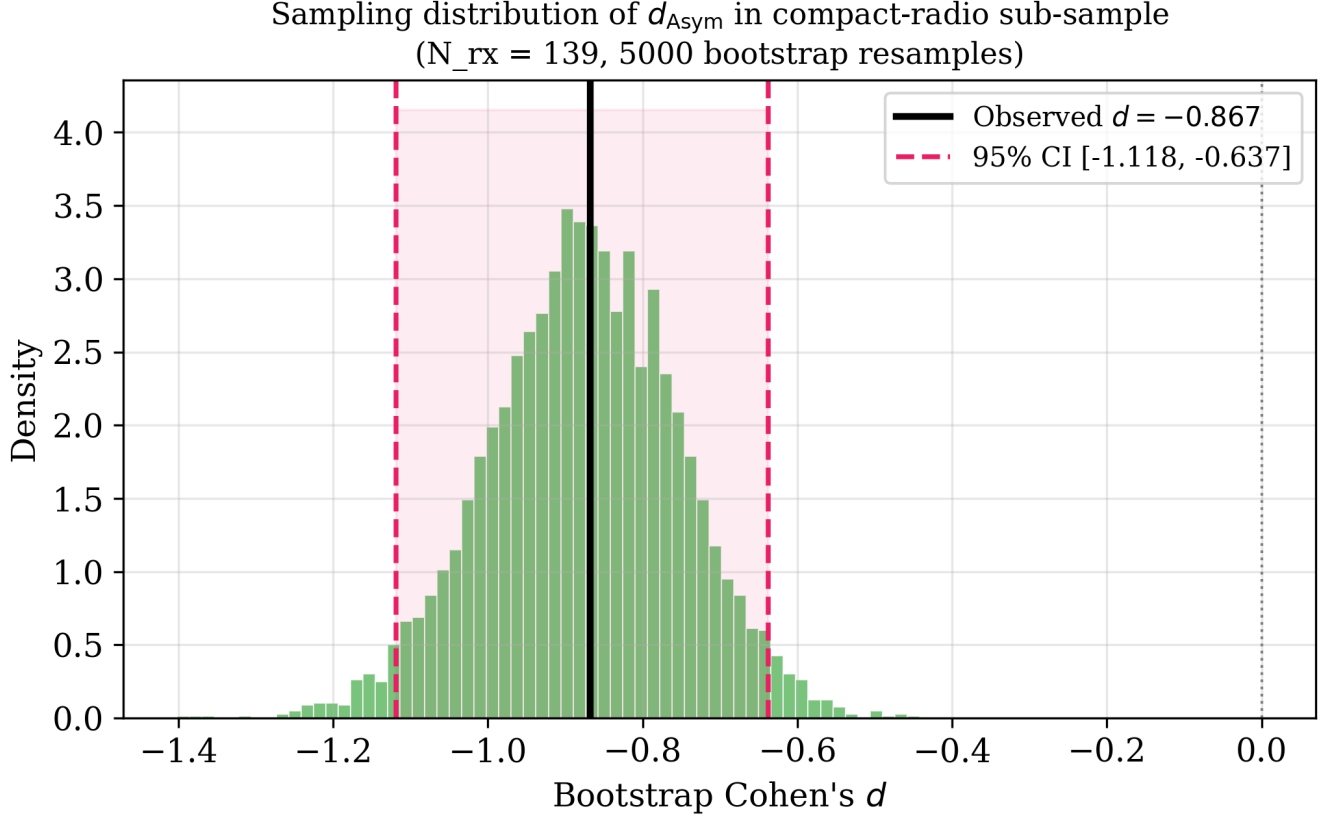


Figure 6. Bootstrap sampling distribution of Cohen’s d for asymmetry in the flat-spectrum matched sub-sample ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$ unique matched pairs; 5000 resamples). The observed $d = -0.867$ (solid vertical line) and 95% CI $[-1.118, -0.637]$ (dashed lines) lie entirely below zero. All 5000 resampled d values are negative, confirming the asymmetry suppression is not a sampling artefact. A Cohen’s d of -0.87 exceeds the conventional large-effect threshold $|d| > 0.8$.

The second channel is *AGN identification*. Asymmetry identifies a specific sub-population: flat-spectrum radio-excess sources hosted by morphologically undisturbed galaxies. These are the primary outliers below the IRRC locus, and their systematically low asymmetry makes them recoverable from the residuals of any radio-only SFR model. Adding morphology improves SFR inference $3.3\times$ more for flat-spectrum sources (0.073 dex residual improvement) than for steep-spectrum ones (0.022 dex; $p = 1.1 \times 10^{-6}$), directly confirming this identification channel.

5.3. Physical Interpretation of the Flat-Spectrum Population

The extreme asymmetry suppression ($d = -0.87$; $p = 8 \times 10^{-13}$) in flat-spectrum radio-excess hosts is the most novel result here. The median asymmetry of these hosts ($\langle A \rangle = 0.059$) is barely half that of their individually matched controls ($\langle A \rangle = 0.110$), and the suppression persists through $\geq 2\sigma_\alpha$ threshold cuts.

One physically consistent interpretation is secular fuelling: bar-driven inflows, minor interactions too subtle to imprint morphological asymmetry, or direct cold-mode accretion, rather than gas-rich major mergers (Hardcastle & Croston 2020). This is consistent with studies of confirmed GPS/CSS host galaxies, which are predominantly early-type morphologically regular systems with little evidence of merger activity (Holt 2009; O’Dea & Saikia 2021). The positive concentration offsets ($d_{\text{CONC}} = +0.35$) reinforce this: compact stellar distributions provide the deep potential required to sustain nuclear activity. Alternative interpretations cannot be excluded; confirmation requires resolved radio morphology from VLBI or e-MERLIN and optical integral-field spectroscopy to constrain Eddington ratios and fuelling mode.

5.4. Parametric Morphology Is Insufficient

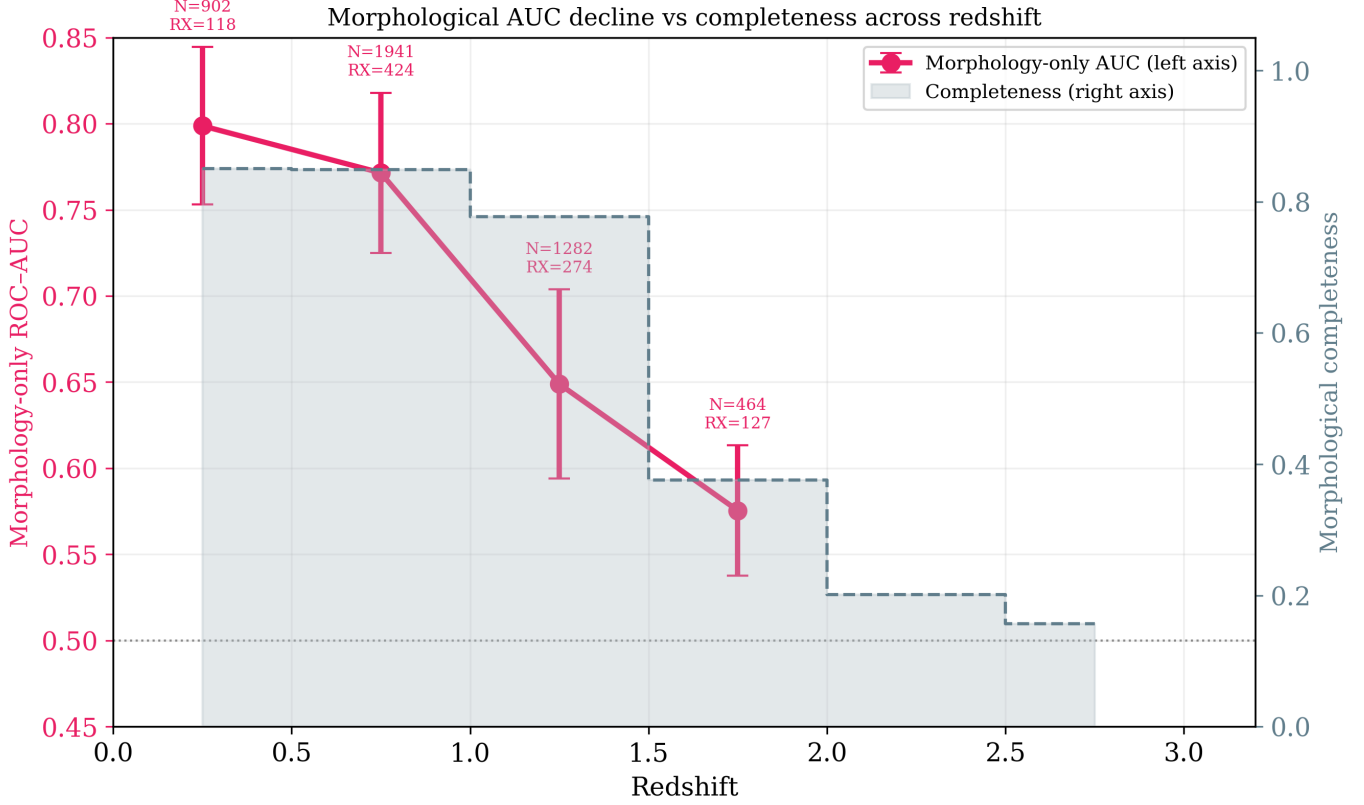


Figure 7. Morphology-only ROC-AUC (left axis, pink circles with error bars) and morphological cross-match completeness (right axis, gray shading) as a function of redshift in $\Delta z = 0.5$ bins. AUC declines from 0.799 ± 0.046 at $z < 0.5$ to 0.575 ± 0.038 at $1.5 < z < 2.0$; completeness declines from $\sim 85\%$ to $\sim 38\%$ over the same interval. Both quantities decline monotonically across all four measured bins, consistent with observational incompleteness driving the AUC decline rather than structural evolution. Numbers annotated above each point give total N (top) and N_{rx} (bottom). The horizontal dotted line marks $AUC = 0.5$ (chance).

A parallel analysis using parametric GALFIT Sérsic morphology (van der Wel et al. 2014) for the 302 VLA-COSMOS sources with available decompositions yields $\Delta R^2 = 0.004$ with no significant matched-sample offsets (all $|d| \leq 0.16$, $p > 0.29$). The smaller sample ($N = 302$ vs. $N = 4843$) reduces statistical power, so the absence of a signal could in principle reflect sample size rather than the nature of the measurements. However, even scaling the non-parametric result to the same N by bootstrap sub-sampling, the Zurich morphology rung consistently delivers $\Delta R^2 > 0.05$ — an order of magnitude larger than the Sérsic result. Sérsic index and effective radius describe only the azimuthally averaged, smooth component of the light profile, and are insensitive to precisely the substructure that drives the radio-morphology signal: nuclear light concentration (Gini), fine-grained radial structure (concentration index), and rotational asymmetry. The factor-of-32 contrast in ΔR^2 is not a sample-size artefact; it reflects a genuine measurement distinction. Future surveys should prioritise non-parametric or citizen-science morphological measurements over Sérsic decompositions for radio-SFR calibration.

5.5. Survey Implications for the SKA Era

Three practical conclusions follow. First, optical morphology from Euclid, Rubin LSST, or Roman will deliver the largest single improvement to radio-SFR calibration: $\Delta R^2 \simeq 0.12$ – 0.16 beyond radio luminosity and stellar mass. After redshift control the residual structural information is $\Delta R^2 = 0.038$ for the Zurich non-parametric metrics and $\Delta R^2 = 0.078$ for the Galaxy Zoo Hubble debiased featured/disk fraction — the latter being the larger of the two because volunteer debiasing has pre-removed the dominant redshift trend. Makechemu et al. (2026) establish that modelling flexibility alone cannot close this gap; morphological information is necessary.

Second, the Galaxy Zoo Hubble result establishes that smooth/featured classification is sufficient. Automated classifiers at citizen-science quality — already demonstrated at scale (Dieleman et al. 2015) — are the right tool; full non-parametric pipelines are not required to achieve the dominant calibration improvement.

Third, flat-spectrum radio-excess sources with smooth morphologies and high concentration form a physically distinct class and should be treated separately in any morphology-based radio survey classification scheme.

6. CONCLUSIONS

We have quantified the hierarchy of host-galaxy information governing radio emission as a star-formation tracer across two independent morphological measurement systems applied to $N = 4843$ VLA-COSMOS 3 GHz sources ($0 < z \lesssim 3$; median $z = 0.91$). Our conclusions are:

1. **Non-parametric morphology is the dominant hidden variable, confirmed by two independent systems.** Gini, concentration, and asymmetry from HST/ACS yield $\Delta R^2 = +0.127$ (MLP), $4.3\times$ the combined gain from all other variables. The Galaxy Zoo Hubble debiased featured/disk fraction yields $\Delta R^2 = +0.156$ from a single volunteer classification. The combined AUC from both systems (0.693 ± 0.009 and 0.697 ± 0.009) are indistinguishable. The convergence of two instruments, two algorithms, and two classifier populations on the same answer confirms the signal is a real physical property of the galaxies.
2. **Genuine structural contributions survive redshift control in both systems, with the citizen-science residual larger in absolute terms.** The Zurich retention fraction is 0.30: $\Delta R^2 = 0.038$ is genuine structural information independent of cosmic evolution. The Galaxy Zoo Hubble retention fraction is 0.50, yielding a redshift-controlled residual of $\Delta R^2 = 0.078$ — more than twice the Zurich value. The higher GZ Hubble retention reflects the debiasing correction (Bamford et al. 2009), which removes the dominant redshift trend from volunteer vote fractions before the ladder is run, leaving a morphological residual that is more genuinely structural than the raw pixel-level metrics.
3. **Radio-excess hosts are compact and morphologically undisturbed at fixed mass, redshift, and SFR.** Zurich: Gini $d = +0.22$ (concentration $d = +0.18$; asymmetry $d = -0.14$). Galaxy Zoo Hubble: spiral $d = -0.434$ ($p = 1.6 \times 10^{-10}$), the largest single-feature effect in either system.
4. **Flat-spectrum radio-excess hosts show extreme asymmetry suppression consistent with secular AGN triggering.** $d_{\text{Asym}} = -0.867$; 95% CI $[-1.118, -0.637]$; $p = 8 \times 10^{-13}$. This large effect ($|d| > 0.8$) persists through threshold cuts up to $2\sigma_\alpha$ and is directionally consistent across both morphology systems.
5. **Parametric Sérsic morphology contributes $32\times$ less information** ($\Delta R^2 = 0.004$; no significant matched-sample offsets). Future surveys must prioritise non-parametric or smooth/featured citizen-science measurements.
6. **A single smooth/featured classification from Euclid, Rubin LSST, or Roman will deliver the dominant radio-SFR calibration improvement in the SKA era.** Whether flat-spectrum radio-excess sources are predominantly GPS/CSS-type AGN fuelled by secular processes is a question for VLBI imaging and integral-field spectroscopy.

¹ J.S.M. acknowledges support from the Science and Technology Facilities Council (STFC) studentship [grant number ST/XXXXXXX/1]. We thank the VLA-COSMOS and Zurich/Tasca teams for making their catalogues publicly available, and the Galaxy Zoo volunteer classifiers whose contributions enabled the citizen-science replication in this paper. This research used Astropy (Astropy Collaboration et al. 2013, 2018, 2022), scikit-learn (Baron 2019), and statsmodels.

Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), scikit-learn (Baron 2019), statsmodels

REFERENCES

- | | |
|--|---|
| <p>Abraham, R. G., van den Bergh, S., & Nair, P. 2003, ApJ, 588, 218, doi: 10.1086/373919</p> | <p>Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74</p> |
|--|---|

- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, *AJ*, 156, 123, doi: [10.3847/1538-3881/aabc4f](https://doi.org/10.3847/1538-3881/aabc4f)
- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, *A&A*, 558, A33, doi: [10.1051/0004-6361/201322068](https://doi.org/10.1051/0004-6361/201322068)
- Bamford, S. P., et al. 2009, *MNRAS*, 393, 1324, doi: [10.1111/j.1365-2966.2008.14252.x](https://doi.org/10.1111/j.1365-2966.2008.14252.x)
- Baron, D. 2019, *Frontiers in Astronomy and Space Sciences*, 6, 57, doi: [10.3389/fspas.2019.00057](https://doi.org/10.3389/fspas.2019.00057)
- Bell, E. F. 2003, *ApJ*, 586, 794, doi: [10.1086/367829](https://doi.org/10.1086/367829)
- Cisternas, M., Jahnke, K., Inskip, K. J., et al. 2011, *ApJ*, 726, 57, doi: [10.1088/0004-637X/726/1/57](https://doi.org/10.1088/0004-637X/726/1/57)
- Condon, J. J. 1992, *ARA&A*, 30, 575, doi: [10.1146/annurev.aa.30.090192.003043](https://doi.org/10.1146/annurev.aa.30.090192.003043)
- Conselice, C. J. 2003, *ApJS*, 147, 1, doi: [10.1086/375001](https://doi.org/10.1086/375001)
- Del Moro, A., Alexander, D. M., Mullaney, J. R., et al. 2013, *A&A*, 549, A59, doi: [10.1051/0004-6361/201219880](https://doi.org/10.1051/0004-6361/201219880)
- Delhaize, J., Smolčić, V., Delvecchio, I., et al. 2017, *A&A*, 602, A4, doi: [10.1051/0004-6361/201629430](https://doi.org/10.1051/0004-6361/201629430)
- Delvecchio, I., Smolčić, V., Zamorani, G., et al. 2017, *A&A*, 602, A3, doi: [10.1051/0004-6361/201629367](https://doi.org/10.1051/0004-6361/201629367)
- . 2021, *A&A*, 647, A123, doi: [10.1051/0004-6361/202040113](https://doi.org/10.1051/0004-6361/202040113)
- Dieleman, S., Willett, K. W., & Dambre, J. 2015, *MNRAS*, 450, 1441, doi: [10.1093/mnras/stv632](https://doi.org/10.1093/mnras/stv632)
- Ellison, S. L., Mendel, J. T., Patton, D. R., & Scudder, J. M. 2013, *MNRAS*, 435, 3627, doi: [10.1093/mnras/stt1562](https://doi.org/10.1093/mnras/stt1562)
- Hardcastle, M. J., & Croston, J. H. 2020, *New Astronomy Reviews*, 88, 101539, doi: [10.1016/j.newar.2020.101539](https://doi.org/10.1016/j.newar.2020.101539)
- Helou, G., Soifer, B. T., & Rowan-Robinson, M. 1985, *ApJL*, 298, L7, doi: [10.1086/184556](https://doi.org/10.1086/184556)
- Holt, J. 2009, in *Proceedings of Science: The Low-Frequency Radio Universe*. <https://arxiv.org/abs/0812.2812>
- Lacki, B. C., Thompson, T. A., & Quataert, E. 2010, *ApJ*, 717, 1, doi: [10.1088/0004-637X/717/1/1](https://doi.org/10.1088/0004-637X/717/1/1)
- Laigle, C., McCracken, H. J., Ilbert, O., et al. 2016, *ApJS*, 224, 24, doi: [10.3847/0067-0049/224/2/24](https://doi.org/10.3847/0067-0049/224/2/24)
- Lintott, C. J., et al. 2008, *MNRAS*, 389, 1179, doi: [10.1111/j.1365-2966.2008.13689.x](https://doi.org/10.1111/j.1365-2966.2008.13689.x)
- Lotz, J. M., Jonsson, P., Cox, T. J., & Primack, J. R. 2008, *MNRAS*, 391, 1137, doi: [10.1111/j.1365-2966.2008.14004.x](https://doi.org/10.1111/j.1365-2966.2008.14004.x)
- Lotz, J. M., Primack, J., & Madau, P. 2004, *AJ*, 128, 163, doi: [10.1086/421849](https://doi.org/10.1086/421849)
- Madau, P., & Dickinson, M. 2014, *ARA&A*, 52, 415, doi: [10.1146/annurev-astro-081811-125615](https://doi.org/10.1146/annurev-astro-081811-125615)
- Magnelli, B., Ivison, R. J., Lutz, D., et al. 2015, *A&A*, 573, A45, doi: [10.1051/0004-6361/201424937](https://doi.org/10.1051/0004-6361/201424937)
- Makechemu, J. S., Chibueze, J. O., & Simmons, B. D. 2026, *ApJ*
- Mechtley, M., Jahnke, K., Windhorst, R. A., et al. 2016, *ApJ*, 830, 156, doi: [10.3847/0004-637X/830/2/156](https://doi.org/10.3847/0004-637X/830/2/156)
- Murphy, E. J., Condon, J. J., Schinnerer, E., et al. 2011, *ApJ*, 737, 67, doi: [10.1088/0004-637X/737/2/67](https://doi.org/10.1088/0004-637X/737/2/67)
- Novak, M., Smolčić, V., Delhaize, J., et al. 2017, *A&A*, 602, A5, doi: [10.1051/0004-6361/201629436](https://doi.org/10.1051/0004-6361/201629436)
- O'Dea, C. P., & Saikia, D. J. 2021, *A&A Rev.*, 29, 3, doi: [10.1007/s00159-021-00131-w](https://doi.org/10.1007/s00159-021-00131-w)
- Peng, C. Y., Ho, L. C., Impey, C. D., & Rix, H.-W. 2002, *AJ*, 124, 266, doi: [10.1086/340952](https://doi.org/10.1086/340952)
- Planck Collaboration, Aghanim, N., et al. 2020, *A&A*, 641, A6, doi: [10.1051/0004-6361/201833910](https://doi.org/10.1051/0004-6361/201833910)
- Schleicher, D. R. G., & Beck, R. 2013, *A&A*, 556, A142, doi: [10.1051/0004-6361/201321707](https://doi.org/10.1051/0004-6361/201321707)
- Schober, J., Schleicher, D. R. G., & Klessen, R. S. 2016, *ApJ*, 827, 109, doi: [10.3847/0004-637X/827/2/109](https://doi.org/10.3847/0004-637X/827/2/109)
- Scoville, N., Aussel, H., Benson, A., et al. 2007, *ApJS*, 172, 1, doi: [10.1086/516585](https://doi.org/10.1086/516585)
- Sérsic, J. L. 1968, *Observatorio Astronomico, Cordoba*
- Smolčić, V., Novak, M., Bondi, M., et al. 2017, *A&A*, 602, A1, doi: [10.1051/0004-6361/201628704](https://doi.org/10.1051/0004-6361/201628704)
- Tasca, L. A. M., Kneib, J.-P., Iovino, A., et al. 2009, *A&A*, 503, 379, doi: [10.1051/0004-6361/200811538](https://doi.org/10.1051/0004-6361/200811538)
- Thompson, T. A., Quataert, E., & Murray, N. 2006, *ApJ*, 645, 186, doi: [10.1086/504035](https://doi.org/10.1086/504035)
- van der Wel, A., Franx, M., van Dokkum, P. G., et al. 2014, *ApJ*, 788, 28, doi: [10.1088/0004-637X/788/1/28](https://doi.org/10.1088/0004-637X/788/1/28)
- Völk, H. J. 1989, *A&A*, 218, 67
- Willett, K. W., et al. 2013, *MNRAS*, 435, 2835, doi: [10.1093/mnras/stt1458](https://doi.org/10.1093/mnras/stt1458)
- . 2017, *MNRAS*, 464, 4176, doi: [10.1093/mnras/stw2568](https://doi.org/10.1093/mnras/stw2568)
- Yun, M. S., Reddy, N. A., & Condon, J. J. 2001, *ApJ*, 554, 803, doi: [10.1086/321380](https://doi.org/10.1086/321380)